

LAB 2

TITLE: Implementation of FCFS (First-Come, First-Served) Scheduling Algorithm in C.

OBJECTIVE: To implement the First-Come First-Served (FCFS) CPU scheduling algorithm and calculate:

- Waiting Time (WT)
- Turnaround Time (TAT)
- Average Waiting Time
- Average Turnaround Time

THEORY:

FCFS is the simplest CPU scheduling algorithm. In this method, the process that arrives first is executed first. It's a non-preemptive algorithm.

♦ Characteristics:

- Processes are scheduled in the order of arrival.
- It may suffer from the convoy effect (i.e., short processes waiting behind long ones).
- Easy to implement and understand.

♦ Terminologies:

- Burst Time (BT): Time required by a process to complete execution.
- Waiting Time (WT): Total time a process has to wait before getting CPU.

$$\circ \text{WT}[i] = \sum \text{BT of all previous processes}$$

- Turnaround Time (TAT): Total time taken from arrival to completion.

$$\circ \text{TAT}[i] = \text{WT}[i] + \text{BT}[i]$$

- Average Waiting Time (AWT):

$$\circ \text{AWT} = \sum \text{WT} / n$$

- Average Turnaround Time (ATAT):

$$\circ \text{ATAT} = \sum \text{TAT} / n$$

Algorithm:

1. Start the program
2. Input number of processes **n**
3. Input burst time for all **n** processes
4. Set **WT[0] = 0**
5. Calculate waiting time for each process:
 - **WT[i] = WT[i-1] + BT[i-1]**
6. Calculate turnaround time for each process:
 - **TAT[i] = WT[i] + BT[i]**
7. Calculate average WT and TAT
8. Display results
9. End

PROGRAMS

```
#include<stdio.h>
```

```
int main() {
    int n, i, j;
    float bt[20], wt[20], tat[20];
    float avwt = 0, avtat = 0;

    printf("Enter total number of processes: ");
    scanf("%d", &n);

    printf("\nEnter Process Burst Time\n");
    for(i = 0; i < n; i++) {
```

```
        printf("P[%d]: ", i+1);
        scanf("%f", &bt[i]);
    }

    wt[0] = 0;
    for(i = 1; i < n; i++) {
        wt[i] = 0;
        for(j = 0; j < i; j++)
            wt[i] += bt[j];
    }
```

```

printf("\nProcess\tBurst Time\tWaiting
Time\tTurnaround Time");
for(i = 0; i < n; i++) {
    tat[i] = bt[i] + wt[i];
    avwt += wt[i];
    avtat += tat[i];
    printf("\nP[%d]\t%.2f\t%.2f\t%.2f", i+1,
bt[i], wt[i], tat[i]);
}

```

```

    avwt /= n;
    avtat /= n;

    printf("\n\nAverage Waiting Time: %.2f",
avwt);
    printf("\n\nAverage Turnaround Time: %.2f",
avtat);

    return 0;
}

```

Output:

```

Enter total number of processes: 3

Enter Process Burst Time
P[1]: 4
P[2]: 5
P[3]: 2

Process      Burst Time    Waiting Time    Turnaround Time
P[1]          4.00          0.00            4.00
P[2]          5.00          4.00            9.00
P[3]          2.00          9.00           11.00

Average Waiting Time: 4.33
Average Turnaround Time: 8.00

```

RESULTS:

The FCFS scheduling algorithm was implemented successfully. The program calculated the **waiting time**, **turnaround time**, and their averages for all processes.

CONCLUSION:

FCFS is a basic CPU scheduling technique. This lab helped in understanding:

- How processes are handled in the order of arrival
- How to calculate process scheduling metrics like WT and TAT